

Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES
Topic 02
Simple Linear Regression

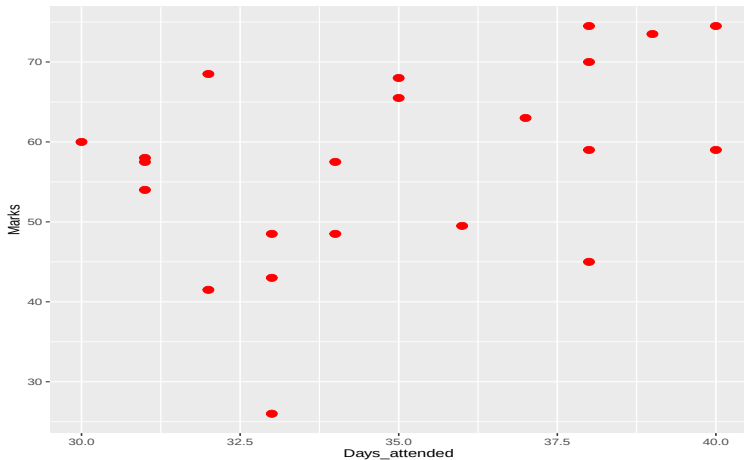
Jan-May 2026

Research Question: What is the impact of attending classes on students' final marks

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Let's start with a real data from IITG which you can feel about it!!

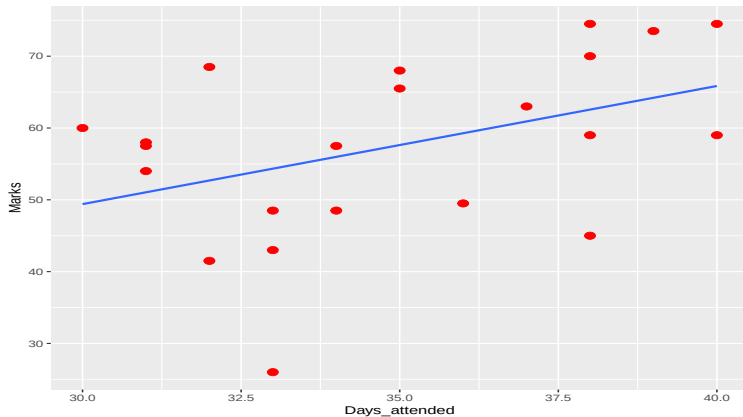
Scatter plot of number of days attended and marks by the students in MTech Data Science course (MA589)



- We will do a regression analysis to the above data to find the answer. Is using regression appropriate here?
- **Strong Advice:** As a Data Scientist/Statistician, you have to **spent a lot of your time** and effort to **pre-process/clean the raw data** to make it **analysis ready**. It is part of your job to clean the data: so learn it quickly!!
- Here is the output after fitting a regression line in R:

Fitting Regression line

Scatter plot of number of days attended and marks by the students in MTech Data Science course (MA589):
with fitted regression line



Linear Models: Simple and Multiple Linear Regressions

The regression framework can be characterized in the following way:

- We have one particular variable that we are **interested in understanding or modeling**, such as sales of a particular product, sale price of a home, or voting preference of a particular voter. This variable is called the **target, response, or dependent variable, and is usually represented** by y .
- We have a set of p **other variables** that we think might be **useful in predicting or modeling the target variable** (the price of the product, the competitor's price, and so on; or the lot size, number of bedrooms, number of bathrooms of the home, and so on; or the gender, age, income, party membership of the voter, and so on). These are called the **predicting, or independent variables, and are usually represented** by $x_1, x_2, \dots, x_p$.

Our Aim

- To build a relationship between y and x .
- That means to write y as a function of x .
- Observed data: (x_i, y_i) for $i = 1, 2, \dots, n$
- We want a function of x , say $f(\cdot)$, such that

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

is as small as possible.

- There are infinite solutions for this problem.
- Therefore, we want to impose constraint the set of functions over which the minimization is performed.

- Typically, a regression analysis is used for one (or more) of three purposes:

- 1 **modeling the relationship** between x and y ;
- 2 **prediction of the target** variable (forecasting);
- 3 and **testing of hypotheses**.

Linear Models: Meaning of Linearity

What does “Linear” mean?

- A linear model is **linear in the parameters β** , but not necessarily in the x 's, e.g.
 - * $y = \beta_0 + \beta_1 x + \beta_2 x^2$ is a **linear model**, because it is linear in β (even though not in x).
 - * $y = \beta_0 + \beta_1 x^{\beta_2}$ is a **non-linear model**, because it is not linear in β .

Transformation

- Clearly $y = f(x) = \beta_0 x^{\beta_1}$ is **not a linear model**, but

$$\ln f(x) = \ln \beta_0 + \beta_1 \ln x$$

If we let $f^* = \ln f$, $\beta_0^* = \ln \beta_0$, $\beta_1^* = \beta_1$, $x^* = \ln x$, we have

$$f^* = \beta_0^* + \beta_1^* x^*,$$

which is **a linear model**.

Transformation

- Thus, although linear models seem to be simple and restrictive, they **can actually be quite flexible by transformation of the response and the predictors.**
- Linear models are **not just straight lines, they can be curved.** Can you give an examples?

Transform the following **non-linear models to linear models**:

- $y = \frac{e^{\beta x}}{1+e^{\beta x}}$, where $y \in (0, 1)$.
- $y = \frac{1}{\beta_0 + \beta_1 x_1 + \beta_2 x_2}.$

Simple Linear Regression

- Just one predictor x , i.e. $p = 1$, where p indicates the number of predictors in a multiple linear regression model.
- The model for the **simple linear regression** is given by

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where y is the **outcome** variable (random), x is the **independent/predictor** variable (non-random) and ϵ is the **random error term**. β_0 (intercept) and β_1 (slope) are model parameters (unknown constants).

- Equivalently, the **model can be written** for $i = 1, 2, \dots, n$ number of observations $(x_1, y_1), \dots, (x_n, y_n)$ as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, i = 1, \dots, n.$$

- How do you **interpret** β_0 (intercept) and β_1 (slope)?

Assumptions:

- 1 The **expected value of the errors is zero**: $E(\epsilon_i) = 0$ for all i
- 2 The **variance of the error is constant** (*homoscedasticity*): $Var(\epsilon_i) = \sigma^2$ for all i
- 3 The **errors** (ϵ_i) are **uncorrelated** with each other: $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$

Assumptions: What they means?

- 1 The **errors** (ϵ_i) are **uncorrelated** with each other. This violation most often occurs in data that are ordered in time (time series data), where errors that are near each other in time are often similar to each other (such time-related correlation is called autocorrelation). **Violation of this assumption** can lead to very **misleading assessments** of the strength of the regression.
- 2 The **expected value of the errors is zero** ($E(\epsilon_i) = 0$ for all i). That is, it cannot be true that for certain observations the model is systematically too low, while for others it is systematically too high.
- 3 The assumption of **homoscedasticity (constant variance)**: each outcome observation (Y) has exact same variance (also for error). If Y follows Poisson distribution, this assumption is violated. Why?

Least Squares Estimation

- Goal: To **estimate** β_0, β_1 **by minimizing error** in some sense (e.g. squared error)
- One reasonable way is to use the ***principle of Least Squares***, i.e. **minimize the objective function**

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

with respect to β_0, β_1 .

- **Differentiate** $Q(\beta_0, \beta_1)$ with respect to β_0, β_1 and **equate the partial derivatives to zero** to get the estimates $\hat{\beta}_0, \hat{\beta}_1$.
- The resulting equations are called ***normal equations***:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

Least Squares Estimation

- The **solution is given by**

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= S_{xy}/S_{xx}\end{aligned}$$

where

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \text{ and } S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are called the **least squares estimator (LSE)** of β_0 and β_1 respectively.

Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are linear combination of y_i 's.
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased for β_0 and β_1 , respectively.
- Variances of $\hat{\beta}_0$ and $\hat{\beta}_1$ are

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \\ \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{S_{xx}} \end{aligned}$$

Linear Estimator

Def: [Linear Estimator] An estimator $\hat{\theta}$ is called linear estimator of θ if $\hat{\theta}$ is a **linear combination** of random observations.

Def: [BLUE] An estimator $\hat{\theta}$ is called the best linear unbiased estimator (**BLUE**) of θ if $\hat{\theta}$ is **linear and unbiased estimator** of θ and $\hat{\theta}$ has **minimum variance among** all linear unbiased estimator of θ .

Gauss-Markov Theorem (Without proof)

Theorem

Under the three assumptions (on slide 11), the LSEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are BLUE of β_0 and β_1 , respectively.

A Few Definitions

- **Fitted values:** $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, $i = 1, \dots, n$
- **Residuals:** Difference between observed and fitted values of y , i.e.

$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n$$

- The objective function evaluated at the LSE, i.e. $Q(\hat{\beta}_0, \hat{\beta}_1)$ is called the **residual sum of squares** (SS_{Res}).

$$SS_{Res} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Properties of Least Squares Fit:

① $\sum_{i=1}^n e_i = 0$. (from the first normal equation)

② $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$. (using (1))

③ $\sum_{i=1}^n x_i e_i = 0$.

④ $\sum_{i=1}^n \hat{y}_i e_i = 0$. (using (1) and (3))

Estimation of Error Variance

- It can be shown that

$$E(SS_{Res}) = (n - 2)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2}$ is called **Residual variance**. Its **square root is called residual standard error**.
- Note that SS_{Res} depends on the model. Therefore, any violation of model assumptions has serious effect on $\hat{\sigma}^2$.
- A convenient computing formula for SS_{Res} is

$$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy},$$

where

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Hypothesis Testing: β_1

- The fourth assumption of linear regression is required for testing of hypothesis: Errors are independent and normally distributed.
- Want to test the hypothesis that the **slope** parameter (β_1) equals to a constant (a value, say β_{10}):

$$H_0 : \beta_1 = \beta_{10} \text{ ag. } H_1 : \beta_1 \neq \beta_{10}$$

- Note that, $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ and y_i 's are independent.
- $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}}) \implies z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi^2_{n-2}$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res}/S_{xx}}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Hypothesis Testing: β_0

- Want to test the hypothesis that the **intercept** parameter (β_0) equals to a constant (a value, say β_{00}):

$$H_0 : \beta_0 = \beta_{00} \text{ ag. } H_1 : \beta_0 \neq \beta_{00}$$

- $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})) \implies z = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{Res}(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Interval Estimation: β_0 and β_1

- To get the CI for β_0 and β_1 , the pivots are

$$\frac{\hat{\beta}_0 - \beta_0}{\sqrt{MS_{Res}\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}}, \text{ and } \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MS_{Res}/S_{xx}}}, \text{ respectively.}$$

- A $100(1 - \alpha)\%$ CI for β_0 is

$$\left[\hat{\beta}_0 \pm t_{n-2, \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} \right].$$

- A $100(1 - \alpha)\%$ CI for β_1 is

$$\left[\hat{\beta}_1 \pm t_{n-2, \alpha/2} \sqrt{\frac{MS_{Res}}{S_{xx}}} \right].$$

Interval Estimation: CI for σ^2

- To get the CI for σ^2 , the pivots is

$$\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$$

- A $100(1 - \alpha)\%$ CI for σ^2 is

$$\left[\frac{(n-2)MS_{Res}}{\chi_{n-2;\alpha/2}^2}, \frac{(n-2)MS_{Res}}{\chi_{n-2;1-\alpha/2}^2} \right].$$

Interval Estimation: CI for mean response

- A regression model can be used to **estimate the mean response** $E(y)$ for a particular value of the regressor variable x . Let x_0 be a value of the regressor variable (must be with the range of original data on x). Then $E(y|x_0) = \beta_0 + \beta_1 x_0$.
- Then, $\hat{y}_0 = \widehat{E(y|x_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0$
- And, $\hat{y}_0 \sim N\left(\beta_0 + \beta_1 x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)$
- Pivot: $\frac{\hat{y}_0 - (\beta_0 + \beta_1 x_0)}{\sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}} \sim t_{n-2}$
- A $100(1 - \alpha)\%$ CI for $\beta_0 + \beta_1 x_0$ is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Prediction Interval for New Observation:

- Let x_0 be a value of the regressor variable (must be with the range of original data on x).
- The true value of the response is y_0 (corresponding to x_0).
- We want to provide an interval I such that $P(y_0 \in I) = 1 - \alpha$
- Note that the point estimate of y_0 is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
- Consider $\psi = y_0 - \hat{y}_0$.
- Then, $E(\psi) = 0, Var(\psi) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$
- $\psi \sim N \left(0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \right)$
- Pivot:
$$\frac{y_0 - \hat{y}_0}{\sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}} \sim t_{n-2}$$

Prediction Interval for New Observation:

- A $100(1 - \alpha)\%$ prediction interval is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Coefficient of determination: R^2

- **Coefficient of determination:**

$$R^2 = \frac{SS_{Reg}}{SS_T} = 1 - \frac{SS_{Res}}{SS_T} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where $SS_{Reg} = \sum_i (\hat{y}_i - \bar{y})^2$

- It is a bounded quantity: $0 \leq R^2 \leq 1$
- R^2 is interpreted as the “**percentage of variation explained by the model**”.
- **Higher values of R^2 are desirable** (R^2 close to 1 indicates a good fit), but “how high is high” **depends on the context**.
- In **multiple linear regression, adding a variable to a model can increase the value of R^2** . To overcome this problem:

$$\text{Adjusted } R^2 : \quad R_{adj}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2 / (n - p - 1)}{\sum_i (y_i - \bar{y})^2 / (n - 1)}.$$

F -Test for Regression

- Large value of the sum of squares for regression $SS_{Reg} = SS_T - SS_{Res}$ indicates the simple regression mean function $E(Y|X = x) = \beta_0 + \beta_1 x$ should be a **significant improvement over the mean function** $E(y|X = x) = \beta_0$.
- This is equivalent to saying that the **additional parameter** (β_1) in the simple regression mean function is **non-zero**. In other words, that $E(Y|X = x)$ is not constant with respect to X .
- To test

$$H_0 : E(Y|X = x) = \beta_0 \text{ ag. } H_1 : E(Y|X = x) = \beta_0 + \beta_1 x$$

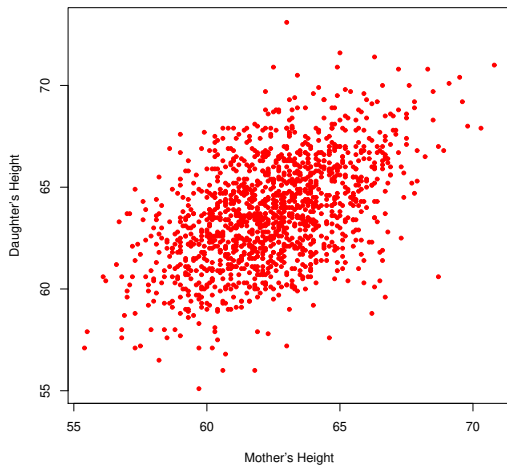
- Test statistics is a ratio, defined as F :

$$F = \frac{SS_{Reg}/1}{\hat{\sigma}^2} = \frac{SS_{Reg}/1}{SS_{Res}/(n-2)} \sim F_{1,n-2}$$

Simple Linear Regression in Heights Data

Heights Data Example

- Data on **heights of** $n = 1375$ **mothers** in the UK under the age of 65 and one of their **adult daughters over the age of 18** (collected and organized during the period 1893–1898 by the famous statistician Karl Pearson).
- A historical **use of regression to study inheritance of height from generation to generation.**



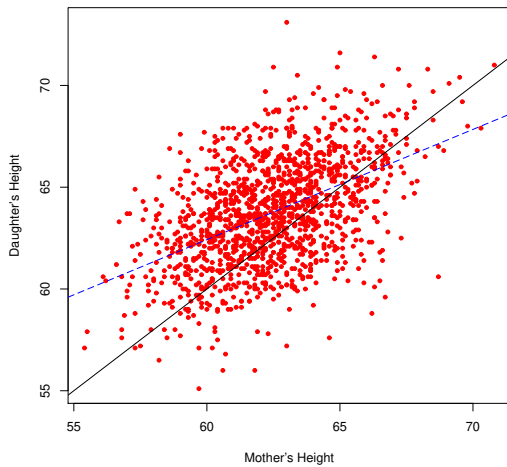
Simple Linear Regression: Using R

```
> height.lm <- lm(Dheight~Mheight,data=heights)
> summary(height.lm)
...
Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
(Intercept) 29.91744    1.62247   18.44  <2e-16 ***
Mheight      0.54175    0.02596   20.87  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.266 on 1373 degrees of
freedom Multiple R-squared: 0.2408,
adjusted R-squared: 0.2402
F-statistic: 435.5 on 1 and 1373 DF,
p-value: < 2.2e-16
```

Let's look at the Regression Line

```
> plot(Dheight ~ Mheight, heights,  
xlab="Mother's Height", ylab="Daughter's Height",  
pch=20, col=2)  
  
> abline(coef(height.lm), lty=5, col=4)  
  
> abline(0,1)
```



Regression Effect

- It's an empirical phenomenon, also called “regression to the mean” or “regression to mediocrity”, e.g.
 - Daughters of tall mothers **tend to be tall, but not as tall** as their mothers; daughters of short mothers tend to be short, but not as short as their mothers (same trend between sons and fathers)
 - Students **doing well in Mid-term tend to do well in the Final, but not as well** as the Mid-term; students doing poorly in Mid-term tend to do poorly in the Final, but not as poorly as the Mid-term!

How to attack the data analysis/model fitting:

- Understand the **research question(s)**. Understand the **data** you have.
 - 1 How the **data was collected**?
 - 2 What type of the **study design used**: *Randomized or Observational; Prospective or Retrospective etc.*
 - 3 Can you make **connection with the primary research question and the data**? Is the **research question feasible** based on the data you have?
 - 4 Are there **secondary research questions**?
 - 5 What are the **potential source of bias**? Sample (data) **may not be** a representative of the target (source) population.
 - 6 Are there any **confounders**? **Confounders**: A confounder (also known as confounding variable, confounding factor) is **a variable that influences both the dependent variable and independent variable**, causing a spurious association. **Confounding is a causal concept**, and as such, cannot be described in terms of correlations or associations.
 - 7 Are the **number of observations/individuals** in the data **sufficient**? Is it a **small data**?

How to attack the data analysis/model fitting:

- Do the **scatter plot(s)**: response vs. input variable(s).
- **Fit the regression** model(s) (or other type of model(s)).
- **Interpret** the output from the fitted models:
 - ① Are all the results expected? Whether the **results go well with existing domain** (basic science) knowledge?
 - ② If not, what are the **reasons behind the aberration** from the expected results.
- **Check the diagnostics** for model assumptions. If you find problem, go back and correct (if you can) the chosen model; or, take decision about the outliers/influential points.
- Always remember: *“Essentially, all models are wrong, but some are useful” – Box*